

Module 5 Neoplasia

• Nomenclature of Neoplasia

All tumor consist of transformed cell(parenchyma) and non-neoplastic supporting cell & BV(stroma)

The nomenclature of tumors is, however based on **the parenchymal component**.

- Benign tumors: attach the suffix –oma to the cell of origin.
- Mesenchymal tumors same nomenclature as benign.
- Epithelial tumors-some based on cells of origin, others on microscopic architecture, and others on their macroscopic patterns.

Exception

- Sound benign but malignant-lymphoma (no benign tumor of lymphocyte),plasmacytoma, glioma, melanoma (benign-Nevus/mole),etc.
- Sound malignant but benign -osteoblastoma, chondroblastoma, etc.

- Malignant Tumors:arising from Mesenchymal tissue - sarcomas

Epithelial cell originfrom any of the three germ layers-carcinomas.

Mixedtumor- divergent differentiation of a single line of parenchymal cells.

- Teratomas: arise **from totipotent**cells that differentiate along various germ lines.
- Blastemal Tumors: **frompluripotential**cells.
- Neoplasm whose cell of origin is unknown(HD,KS,ES)

Tumor- like conditions

- Hamartoma- mass of disorganized tissue indigenou to the particular site.
- Choristoma -is non-neoplastic, tumor like masses of ectopic tissue.

• Benign an malignant neoplasms; differentiation, anaplasia

Benign & malignant neoplasms;

- Both are monoclonal.
- **Benign-** slow growth rate, no invasion, no metastasis and high patient survival rates after successful surgical removal.
- **Low grade malignant** (locally aggressive,borderline tumor) – variable growth rate, locally invasive, low or no metastatic potential and intermediate patient survival rates; tendency for local recurrence after successful surgical removal.

- **Malignant(cancer)**- rapid growth rate, invasive, metastasizing, poor patient survival rates; tendency for local and distant recurrence(metastasis) after successful surgical removal.

Differentiation & anaplasia:

- Differentiation- the extent to which parenchymal cells resemble comparable normal cells, both morphologically & functionally.
- Anaplasia- **complete lack of differentiation** which is only seen in parenchymal cell that constitute transformed cell.

• **Cancer stem cells**

- Rare cells within tumors with the ability to self-renew and give rise to the phenotypically diverse tumor cell population to drive tumorigenesis.
- Arise from normal stem cell or from more differentiated cells.
- Insensitive to chemotherapy. (*Eg of cancers that are treated with stem cells -Acute Leukemia, Chronic Leukemia*)

Theory of Cancer Stem Cells

- Cancer cells are not all the same.
- Among all cancerous cells, a few act as stem cells.
- Cancer is primarily driven by a smaller population of stem cells.

Two General Models for Cancer Heterogeneity (b/c of high genetic change and mutation)

1. All cancer cells are potential cancer stem cells but have a low probability of proliferation in clonogenic assays.
2. Only a small definable subset of cancer cells are cancer stem cells that have the ability to proliferate indefinitely.

• **Local invasion, Metastasis, pathways of metastasis**

Local Invasion

- ❖ A benign neoplasm remains localized at its site of origin. It does not have the capacity to infiltrate, invade, or metastasize to distant sites, as do malignant neoplasms.
- ❖ *Cancers grow by progressive infiltration, invasion, destruction, and penetration of the surrounding tissue.* They do not develop well-defined capsules. There are, however, occasional instances in which a slowlygrowing malignant tumor deceptively appears to beencased by the stroma of the surrounding host tissue, butmicroscopic examination usually reveals tiny crablike feetpenetrating the margin and infiltrating adjacent structures.The infiltrative mode of growth makes it necessary toremove a wide margin of surrounding normal tissue whensurgical excision of a malignant tumor is attempted.Surgical pathologists carefully examine the margins of resected tumors to ensure that they are devoid of cancer cells (*clean margins*).
- ❖ *Next to the development of metastases,local invasiveness is the most reliable feature that distinguishesmalignant from benign tumors.*

Metastasis

- ❖ *More than any other attribute, the property of metastasis identifies a neoplasm as malignant.* Not all cancers have equivalent ability to metastasize, however. At one extreme are basal cell carcinomas of the skin and most primary tumors of the central nervous system, which are highly invasive locally but rarely metastasize. At the other extreme are osteogenic (bone) sarcomas, which usually have metastasized to the lungs at the time of initial discovery.
- ❖ In general, the more anaplastic and the larger the primary neoplasm, the more likely is metastatic spread, but as with most rules, there are exceptions.
- ❖ Dissemination strongly prejudices, and may preclude, the possibility of curing the disease.
- ❖ Malignant neoplasms disseminate by one of three pathways:
(1) seeding within body cavities, (2) lymphatic spread, or (3) hematogenous spread.
- ❖ *Spread by seeding* occurs when neoplasms invade a natural body cavity. This mode of dissemination is particularly characteristic of **cancers of the ovary**, which often cover the peritoneal surfaces widely. Neoplasms of the central nervous system, such as a **medulloblastoma or ependymoma**, may penetrate the cerebral ventricles and be carried by the cerebrospinal fluid to re-implant on the meningeal surfaces, either within the brain or in the spinal cord.
- ❖ *Lymphatic spread* is more typical of carcinomas, whereas *hematogenous spread* is favored by sarcomas. There are numerous interconnections, however, between the lymphatic and vascular systems, so all forms of cancer may disseminate through either or both systems.

✓ Lymphatic spread

A “sentinel lymph node” is the first regional lymph node that receives lymph flow from a primary tumor. But, in some cases, cancer cells seem to traverse the lymphatic channels within the immediately proximate nodes to be trapped in subsequent lymph nodes, producing so-called *skip metastases*.

Of note, although enlargement of nodes near a primary neoplasm should arouse concern for metastatic spread, it does not always imply cancerous involvement. The necrotic products of the neoplasm and tumor antigens often evoke immunologic responses in the nodes, such as hyperplasia of the follicles (lymphadenitis) and proliferation of macrophages in the subcapsular sinuses (sinus histiocytosis). Thus, histopathologic verification of tumor within an enlarged lymph node is required.

✓ Hematogenous spread

Arteries are penetrated less readily than are veins. Since all portal area drainage flows to the liver, and all caval blood flows to the lungs, **the liver and lungs are the most frequently involved secondary sites in hematogenous dissemination**.

Cancers arising near the vertebral column often embolize through the paravertebral plexus; this pathway probably is involved in the frequent vertebral metastases of carcinomas of the thyroid and prostate.

Certain carcinomas have a propensity to grow within veins. **Renal cell carcinoma** often invades the renal vein to grow in a *snakelike fashion* up the inferior vena cava, sometimes reaching the right side of the heart. Hepatocellular carcinomas often penetrate portal and hepatic radicles to grow within them into the main venous channels

- **Essential Alterations of Malignant Transformation**

There are seven of these

1. **Self-sufficiency in growth signals**- ability to proliferate in the absence of external stimuli and this is achieved by oncogene activation
2. **Insensitivity to growth-inhibitory signals**: tumors are not responsive to normal growth inhibitors like TGF- β and direct inhibitors of cyclin-dependent kinases (CDKs).
3. **Evasion of apoptosis**: resistant to programmed cell death, as a consequence of p53 inactivation or activation of anti-apoptotic genes.
4. **Limitless replicative potential**: can avoid cellular senescence and mitotic catastrophe.
5. **Sustained angiogenesis**: tumors must induce angiogenesis as they need vascular supply to bring nutrients and oxygen and remove waste products.
6. **Ability to invade and metastasize**: metastases depend on processes that are intrinsic to the cell or are initiated by signals from the target tissue environment.
7. **Defects in DNA repair**: DNA repair genes are not directly involved in tumor genesis it is the problem in DNA repair gene regulators that causes it.

- **Growth factors and their functions**

- a type of chemical messenger
- a substance that stimulates cell proliferation and often also promotes cell differentiation of specific target cells
- Growth factors are **polypeptides** that function through stimulation of cellular proliferation. Some growth factors are considered hormones, and some have been called cytokines. Each of the hundreds of chemical messengers has its own specific receptor, which will usually bind no other messenger.
- Many GF are synthesized by Stromal cells and normally sequestered by proteins within ECM.
- Responsible for controlling functions like growth and development, differentiation, proliferation, migration, survival and apoptosis.
- Are synthesized by various cells in the body and act via various routes: **endocrine**- long range dispersion via circulation e.g. IGFs, **paracrine**- release by local sources e.g. TGF β , **autocrine**- cell responds to growth factors that it produces itself e.g. tumor cell
- In cancerous tissue, the secretion of certain of these chemical messengers become unregulated and cellular proliferation continues unabated.
- Some of the growth factors are found on **table 4.4**

- *Growth factors* mediate their diverse biologic responses by binding to and activating cell-surface *receptors* with intrinsic protein kinase activity or intracellular receptors
- Membrane bound receptors for GF are various types
 1. Receptor tyrosine kinases (RTKs): activate a number of downstream effectors and mediate a lot of developmental, cell proliferation, and cell differentiation pathways. Alterations of tyrosine kinase-mediated pathways are often observed in human cancers.
 - Activation of RTK: **binding of ligand** to the extracellular domain causes **dimerization** of two RTK molecule. Then the receptor undergoes intermolecular **auto or trans-autophosphorylation** on **tyrosines**, this further stimulates the kinase activity of the RTK.
 - Dominant-negative inhibitors of RTKs: if a receptor mutant is expressed with inactive kinase, **it will dimerize but will not trans-autophosphorylate**. This will block activation of receptors.
 - PDGF receptor signaling complex: is phosphorylated on several tyrosines.
 - SRC homology region (SH-2) Domain: binding of a phosphotyrosine residues requires either a SH-2 domain or PTB domain. SH-2 domain have two binding sites. One for phosphotyrosine and second that binds specific a.a residue.
 - Activity of phosphoinositide 3-kinases: is a major signaling molecule involved in cell growth. It promotes the cell growth necessary to allow cells to divide and mature. It also promote the survival of multiple cell by inhibiting apoptosis.
 - Activation of RAS : activated by GEFs. Grb-2 protein which is attached to the tyrosine will recruit ras GEF where it exchanges GDP for GTP in Ras and activated Ras. Ras is a critical mediator of RTK signaling.
 2. Non-receptor protein tyrosine kinases (nrTKs): located in the cytoplasm or nucleus. Has no intrinsic tyrosine kinase activity.
 - JAK-STAT Pathway: activated by cytokines, receptor oligomerization leads to activation of the associated JAK kinases and this will provide a docking site for STAT proteins. The stats are phosphorylated by JAKs which causes their dimerization. The dimerized STAT translocate into the nucleus and promote gene transcription.
 3. Receptor serine/ threonine kinase signaling:
 - TGF b family signaling: is an extracellular protein that binds a cell surface receptor. It causes cells to arrest their cell cycle and enter Go.
 - Upon binding to TGF b, the receptor phosphorylates proteins in the cytoplasm called Smads. These enter the nucleus and function as transcription factors to turn on specific target genes.
 - A key gene turned on by TGF b is Cdk inhibitor p21.
- Signaling networks:
 - Signaling diversity: most receptors activate multiple intracellular signaling pathways. So a single receptor can have multiple effect on cells
 - Cross talk: signaling pathways contain points where they can be regulated by other signaling pathways. This enables one signaling pathway to "branch out" and affect other target proteins
 - Redundancy: different signal molecules can have similar effect on cells

- Signal amplification: very small amounts of a ligand can have dramatic effect.
- Platelet- derived growth factor (PDGF): causes migration and proliferation of fibroblasts, smooth muscle cells and monocytes to area of inflammation and healing skin wounds. Bind to PDGFR alpha or beta
- PDGFR- The extracellular region has five ig-like domains that function in ligand binding and receptor dimerization. Grb2, Ptdlns 3-kinase, PLCy, InsP3 and DAG.
- Epidermal Growth Factor: results in cellular proliferation, differentiation and survival.
- EGFR family-EGFR (ErbB), ErbB2,3,4/HER2,3,4 (RESPECTIVELY)
 - RAS/ERK pathway: results in cell proliferation and in the increased transcription of Bcl2 family members and inhibitors of apoptosis proteins (IAPs), thereby promoting cell survival.
 - PI3 Kinase/AKT pathway: AKT promotes cell survival through the transcription of anti-apoptotic proteins.
- EGFR family and cancer: in cancer ErbB2 expression is increased in breast cancer, ovarian, gastric and bladder cancers.
 - some tumors increase the production of EGF-related growth factors leading to the persistent activation of ErbB receptors.
- Therapies used for the treatment of EGF fuelled cancers include:
 - suramin, inhibit binding of EGF to its receptor
 - Gfitinib, prevent autophosphorylation by targeting ErbB1 tyrosine kinase activity
 - Herceptin, monoclonal AB that targets the ErbB2 receptor and triggers receptor internalization, so no signaling.
- Fibroblast Growth Factor family: produced by keratinocytes, fibroblasts, endothelial cells, smooth muscle cells, chondrocytes, and mast cells.
- FGFs can bind to heparin and have their biologic activities modulated, hence they are also termed as heparin-binding growth factor (HBGFs).
- Vascular endothelial growth factor: specific mitogen for vascular endothelial cells.
- VEGF is a potent inducer of blood vessel formation in early development (vasculogenesis) and has a central role in the growth of new blood vessels (angiogenesis) in adults.
 - It promotes angiogenesis in chronic inflammation, healing of wounds and in tumors.
- The Insulin Family: involved in regulation of rapid anabolic responses, such as glucose uptake, lipogenesis and a.a and ion transport. It also stimulates DNA synthesis and cell growth.
- Insulin like growth factors (IGFs): regulate growth, activation and differentiation of cells nd tissues. Synthesized in the liver then go to the bloodstream.
- IGF effect: promote cell cycle progression, organogenesis in fetal development, neuronal tissue growth and differentiation

Table 4–4. Characteristics of Some Representative Growth Factors

Factor	Original Source	Target Cell
Insulin	Beta cells of pancreas	General
Insulin-like growth factors		
IGF-1	Human plasma	General
IGF-2	Human plasma	General
Basic fibroblast growth factor (FGF-2)	Bovine pituitary	Fibroblasts, myoblasts, smooth muscle, chondrocytes, glial cells, vascular endothelium
Nerve growth factor (NGF)	Mouse submaxillary gland, snake venoms, cultured cells	Sympathetic ganglia cells and sensory neurons
Epidermal growth factor (EGF)	Mouse submaxillary gland, human urine	Epidermal cells, various epithelial cells, vascular endothelial cells, chondrocytes, fibroblasts, glial cells
Platelet-derived growth factor (PDGF) AA, AB, and BB	Human platelets	Fibroblasts, glial cells, arterial smooth muscle cells
Transforming growth factors		
TGF- α	Various virally transformed cell types and cancer cells	Similar to EGF
TGF- β	Various transformed cell types, normal placenta, kidney, and platelets	Similar to EGF
Angiogenin	Human colon carcinoma cell line	Capillary endothelium
Colony-stimulating factors (human)		
GM-CSF	Placenta	Granulocyte/macrophage
G-CSF	Placenta	Granulocyte progenitor cells
M-CSF	Urine	Macrophage progenitor cells
Interleukins		
IL-1	Normal and malignant macrophages, keratinocytes, astrocytes	T lymphocytes, B lymphocytes, fibroblasts, hepatocytes, chondrocytes, hypothalamic cells (fever center)
IL-2 (TCGF)	T lymphocytes	T lymphocytes (that become T-helper cells or cytotoxic T cells)
IL-3 (Multi-CSF)	T lymphocytes	Eosinophil, mast cell, granulocyte, and macrophage progenitors; T lymphocytes
Mammary-derived growth factor (MDGF)	Human milk and mammary tumors	Normal mammary epithelial cells, epidermoid carcinoma cells
Uterine-derived growth factor (UDGF)	Pregnant sheep uterus	Rat mammary and uterine tumor cells

• Cell cycle and Carcinogenesis

- ✓ Cell divisions proceed at an amazing clip, with DNA replication beginning immediately after mitosis ends. However, as development proceeds, two gaps are incorporated into the cell cycle: Gap 1 (G_1) between mitosis (M) and DNA replication (S), and Gap 2 (G_2) between DNA replication (S) and mitosis (M) .
- ✓ Although each phase of the cell cycle circuitry is monitored carefully, the transition from G_1 to S is believed to be an extremely important checkpoint in the cell cycle clock. Once cells cross the G_1 checkpoint they can pause the cell cycle for a time, but they are obligated to complete mitosis. In G_1 , however, cells can exit the cell cycle, either temporarily, called quiescence, or permanently, called senescence. In G_1 , therefore,

diverse signals are integrated to determine whether the cell should enter the cell cycle, exit the cell cycle and differentiate, or die.

- ✓ R.b:- cell cycle G1 –S- G2-M
- ✓ Carcinogenesis is a multistep process, and development of all human cancers appears to require the accumulation of multiple genetic changes. These changes can involve either inherited germ-line mutations or acquired mutations. Four classes of normal regulatory genes are the targets of genetic damage:

1.Growth-promoting proto-oncogenes

Protooncogenes are normal cellular genes involved with growth and cellular differentiation. Oncogenes are derived from protooncogenes by either a change in the gene sequence, resulting in a new gene product (oncoprotein), or a loss of gene regulation resulting in overexpression of the normal gene product. Mechanisms of oncogene activation include point mutations, chromosomal translocations, gene amplification and insertional mutagenesis. Activated oncogenes lack regulatory control and are overexpressed, resulting in unregulated cellular proliferation. In oncogene one copy is sufficient to cause cancer unlike tumor suppressor.

Oncogenes can be classified as:

- Growth factors
- Growth factor receptors
- Signal transducers

Ras -A single amino acid change in Ras protein can cause constant stimulation of the pathway, even in the absence of growth factors because they locked into its GTP-binding shape.*presence of ras mutations is detected in several human tumors- 90% of pancreatic, 50% of colon and 30% of lung.*

H- Ras –bladder cancer

K –Ras -lung ,colon and pancreas

- Transcription factors (nuclear regulatory proteins)

70% of cancers have deregulated Myc. Chromosomal translocations increase c-Myc transcription in Burkitt's lymphoma and other lymphoid malignancies.

- Cell cycle regulators

2.Growth-inhibiting tumor-suppressor genes

Tumor suppressor genes encode proteins that regulate and suppress cell proliferation by inhibiting progression of the cell through the cell cycle. The mechanism of action of tumor suppressor genes may vary. As examples, p53 prevents a cell with damaged DNA from entering S-phase, while Rb prevents the cell from entering S-phase until the appropriate growth signals are present.

Knudson's "two hit hypothesis" states that at least 2 tumor suppressor genes must be inactivated for oncogenesis. In cancers arising in individuals with inherited germ-line mutations, the "first hit" is the inherited germ-line mutation and the "second hit" is an acquired somatic mutation. Examples of inherited germ-line mutations include familial

Retinoblastoma in which germ-line mutation of Rb on chromosome 13 is associated with a high rate of retinoblastoma and osteosarcoma.

Li-Fraumeni syndrome in which germ-line mutation of p53 on chromosome 17 is associated with a high rate of many types of tumors. In normal cell p53 protein are low but during DNA damage and other stressful condition p53 level increase resulting DNA repair or growth arrest or induce apoptosis.

3. Genes that regulate apoptosis

Tumor genesis related to changes in regulation of apoptosis occurs in the follicular lymphomas that have the translocation $t(14; 18)$. Normally, bcl-2 prevents apoptosis (programmed cell death). In the follicular lymphomas with this translocation, the bcl-2 regulator of apoptosis is overexpressed, because the translocation connects the immunoglobulin heavy chain gene on chromosome 14 (which turns on easily in B lymphocytes) to the bcl-2 gene on chromosome 18, thereby leading to a situation in which lymphocytes fail to die as expected and instead produce a tumor. Other examples of apoptosis regulators include bax, bad, bcl-xS, and bid; p53 promotes apoptosis in mutated cells by stimulating bax synthesis. c-myc promotes cellular proliferation with bcl and when associated with p53 leads to apoptosis.

4. Genes that regulate DNA repair; defective DNA repair predisposes to genomic mutations which will result tumor development along other mutation.

• Tumor Biomarkers

- ✓ Are oncoproteins (products of mutated oncogene) or mutated forms of oncoproteins that indicate the presence of tumor. Could be related to proliferation, differentiation or metastasis.
- ✓ Hormones like HCG, serum proteins, enzymes and metabolites could also be elevated in non-malignant cases giving false positive results thus are not used for screening but are used in monitoring treatment since the serum level of tumor markers reflect well on the success of a surgery or efficacy of chemotherapy.
- ✓ The protein composition may be associated with disease processes in the organism and thus have potential utility as diagnostic markers as Proteins are closer to the actual disease process, in most cases, even more than parent genes.
- ✓ Cell products produced or released during metastasis indicate the risk of metastasis or poor prognosis e.g. proteases and adhesion molecules can be used to predict prognosis. Most are measured in tumor tissues and cytosols.
- ✓ Carcinoembryonic proteins can be used as markers related to differentiation as they are found in fetal tissue and tumors but not normal adult tissue. Though it has higher sensitivity and specificity than metabolites, it is not used for screening because of its cross reactivity. (Measured in immunoassay due to small concentration)

- ✓ Inherited mutations of suppressor genes can be used for screening and identification of families or individuals at higher risk. Example BRCA1 and BRCA2 for breast cancer, BRCA2 risk of breast cancer in men, BRCA1 for ovarian, colon and prostate cancer.

Examples of tumor biomarkers

- a. CEA- is the most widely used marker for GI cancer
 - Metabolized in liver(impaired clearance in liver damage)
 - Is used to follow patients during therapy to detect recurrence
 - May increase in patients following chemotherapy or radiotherapy
 - b. C-ERB2/ Her-2/Neu- associated with tyrosine protein kinase
 - Amplified in 25-30% of breast and ovarian cancer
 - An important predictor of both disease relapse and survival
 - Superior to all prognostic factors but lymph nodes.
 - Used to identify patients with breast cancer who would most likely be benefited by high dose adjuvant chemotherapy
 - Herceptin is a monoclonal antibody against its receptor, used to treat malignant breast cancer
 - c. ER/PR-to identify patients who are most likely to benefit from endocrine therapy
 - Positivity indicates good prognosis
 - d. P53- found mutated in almost half of all cancers
 - Can be measured in tissue, fibroblast, WBC or serum
 - e. PS2 protein-a peptide secreted from breast cells induced by estrogen.
 - Capable of predicting response to endocrine therapy
 - Negative value is associated with earlier recurrence and death.
 - f. PSA(prostate specific antigen)
 - The best tumor marker due to its tissue specificity
 - For screening and management of prostate cancer
 - Lacks cancer specificity as it is also elevated in BPH, prostatitis and infraction
 - Used to monitor the success of prostatectomy and recurrence
 - g. SCC Ag (squamous cell carcinoma antigen)
 - Elevated in 70% of advanced cervical cancer
 - Highest in patients with metastasis
 - Used in monitoring SCC carcinoma of head, neck, lung, esophagus, anal canal
- ❖ All of the biomarkers have neither the sensitivity nor the specificity to be used for screening so it is not recommended to use them especially in asymptomatic population but can be used for screening in a population where there is high incidence.

• Multistep Carcinogenesis and Cancer Progression and heterogeneity

- ✓ Carcinogenesis is a multistep process at both the phenotypic and the genetic levels, resulting from the accumulation of multiple mutations. Typically about 6-7

independent mutations are required for carcinogenesis (Transfer of a single oncogene to a normal cell is normally not sufficient to transform and neither is the loss of one allele of a TS gene). Cancer frequency increases with age, suggesting that transformation of cells requires the accumulation of multiple mutations.

- ✓ Most oncogenes can induce both growth and apoptosis so there needs to be another that inhibits cell death. This is called oncogene cooperation. Mutations that don't cause cancer just on their own end up causing cancer when paired with another mutation e.g. Ras/p16 BRCA1/p53 p27/Rb Myc/p53 Myc/Ras.
- ✓ Mutator genes increase spontaneous mutation rate of other genes. Its products are involved in DNA replication and repair; mutations make the cell error prone example HNPCC (Hereditary non-polyposis colon cancer) Mutation at any one of 4 genes (hMSH2, hMLH1, hPMS1, hPMS2) leads to predisposition.
- ✓ Cancer doesn't appear overnight it takes time and goes through multiple changes. We can take colorectal carcinoma as an example.

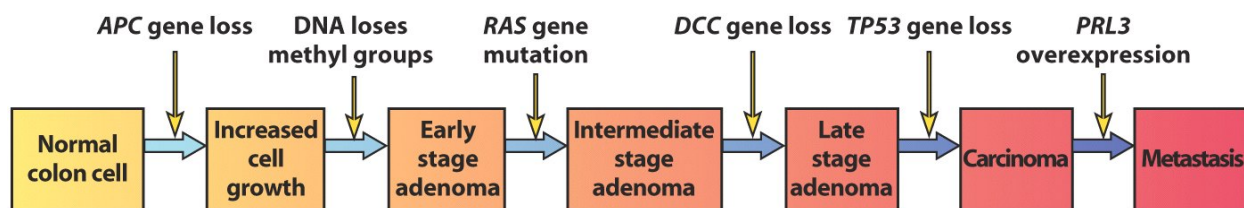


Figure 16-8 Cell and Molecular Biology, 4/e (© 2005 John Wiley & Sons)

- ✓ During progression, tumor cells are subjected to immune and non-immune selection pressures. A growing tumor, therefore, tends to be enriched for subclones that “beat the odds” and are adept at survival, growth, invasion, and metastasis

HETEROGENEITY

- ❖ At the molecular level, tumor progression and associated heterogeneity are most likely to result from multiple mutations that accumulate independently in different cells, generating subclones with different characteristics such as ability to invade, rate of growth, metastatic ability, karyotype, hormonal responsiveness, and susceptibility to antineoplastic drugs. Some of the mutations may be lethal; others may spur cell growth by affecting proto-oncogenes or cancer suppressor genes. Thus even though most malignant tumors are monoclonal in origin, by the time they become clinically evident their constituent cells may be extremely heterogeneous.
- ❖ There are two theories to explain heterogeneity: cancer stem cell model and the clonal evolution model. According to CSC theory, heterogeneity is due to differences in the stem cells from which they originated and difference in CSC is due to epigenetic changes/clonal evolution. But CEM states that tumors arise from a single mutated cell, accumulating additional mutations as it progresses.

PROGRESSION

❖ Accumulation of mutations eventually results in tumor invasion and spread.

1. Epithelial tumor cells are normally attached to one another by cellular adhesion molecules (e.g. E-cadherin).

2. Down regulation of E-cadherin leads to dissociation of attached cells.

3. Cells attach to laminin and destroy basement membrane (collagen type IV) via collagenase.

4. Cells attach to fibronectin in the ECM and spread locally,

5. Entrance into vascular or lymphatic spaces allows for metastasis.

• **Genetic and non-hereditary predisposition to cancer**

- ✓ Approximately 90-95% of all cancers are sporadic only 5-10% are inherited.
- ✓ Germ line mutation in egg or sperm are heritable and cause cancer family syndrome. somatic mutation (e.g. breast) that occur in non germ cell are not heritable.
- ✓ Inheritance could be

1. **Autosomal dominant:** retinoblastoma, familial adenomatous polyposis (colon cancer)

- ✓ Childhood retinoblastoma is the most striking example of this category. Approximately 40% of retinoblastomas are familial. inherited disabling mutations in a *tumor suppressor gene*. (i.e. *RB*) are responsible for the development of this tumor in families. Carriers of this gene have a 10,000-fold increased risk of developing retinoblastoma. Unlike those with sporadic retinoblastoma, patients with familial retinoblastoma develop bilateral tumors, and they also have a greatly increased risk of developing a second cancer, particularly osteosarcoma

2. **Autosomal recessive:** xeroderma pigmentosum in which DNA repair is defective.

3. X-linked recessive: Wiskott-Aldrich syndrome (malignant lymphoma)

4. *Familial Cancers of Uncertain Inheritance*

- ✓ Examples are carcinomas of colon, breast, ovary, and brain. *Features that characterize familial cancers include early age at onset, tumors arising in two or more close relatives of the index case, and sometimes multiple or bilateral tumors.* Familial cancers are not associated with specific marker phenotypes.

❖ Epigenetic Modifications and Cancer

- ✓ *Epigenetics* refers to reversible, heritable changes in gene expression that occur without mutation. Such changes involve posttranslational modifications of histones and DNA methylation, both of which affect gene expression. During the past several years that tumor suppressor genes are sometimes silenced by hypermethylation of promoter sequences, rather than by mutation. CDKN2A is a complex locus that encodes two tumor suppressors, p14/ARF and p16/INK4a, produced from two different reading frames;

p14/ARF is epigenetically silenced in colon and gastric cancers, while p16/INK4a is silenced in a wide variety of cancers. Since this locus produces two tumor suppressors that affect the p53 and Rb pathways, silencing this locus has the pleasing effect (from the cancer's standpoint) of removing two checkpoints with a single alteration.

✓ MicroRNAs and Cancer

miRNAs can participate in neoplastic transformation either by increasing the expression of oncogenes or reducing the expression of tumor suppressor genes. If an miRNA inhibits the translation of an oncogene, a reduction in the quantity or function of that miRNA will lead to overproduction of the oncogene product. Conversely, if the target of a miRNA is a tumor suppressor gene, then overactivity of the miRNA can reduce the tumor suppressor protein. Such relationships have already been established by miRNA profiling of several human tumors. For example, downregulation or deletion of certain miRNAs in some leukemias and lymphomas results in increased expression of *BCL2*, the antiapoptotic gene. Similar miRNA-mediated upregulation of the *RAS* and *MYC* oncogenes also has been detected in lung tumors and in certain B cell leukemias, respectively.

- ✓ Just as some hereditary conditions increase the risk of getting certain cancers, so do certain acquired conditions. precursor lesions arise in the setting of chronic tissue injury or inflammation, which may increase the likelihood of malignancy by stimulating continuing regenerative proliferation or by exposing cells to byproducts of inflammation, both of which can lead to somatic mutations (discussed later). Indeed, molecular analyses have shown that many precursor lesions possess some of the genetic lesions found in their associated cancers. Clinically, these precursor lesions are important to recognize, because their removal or reversal may prevent the development of a cancer.

☒ A brief listing of some of the chief precursor lesions follows:

- Squamous metaplasia and dysplasia of the bronchial mucosa, seen in habitual smokers—a risk factor for lung cancer
- Endometrial hyperplasia and dysplasia, seen in women with unopposed estrogenic stimulation—a risk factor for endometrial carcinoma
- Leukoplakia of the oral cavity, vulva, or penis, which may progress to squamous cell carcinoma
- Villous adenomas of the colon, associated with a high risk of transformation to colorectal carcinoma

• **Tumor immunity**

tumors are not entirely “self” and immune-mediated recognition of autologous tumor cells may be a “positive mechanism” capable of eliminating transformed cells.

- ***immune surveillance hypothesis*** states that immune system plays only a positive role via recognition and destruction of newly appearing tumor cells, which are seen as foreign by the host immune system. But the fact that cancers occur implies that immune surveillance is imperfect; the escape of some tumors from such policing. Therefore, this theory is later-on replaced by ***cancer immune-editing hypothesis***, which implies to the paradoxical

role of the immune system in **preventing and controlling tumor outgrowth at early stages (elimination phase)**, but if that fails, **facilitating cellular transformation via shaping (editing) immunogenicity of tumors at later stages (escape phase)**.

- **Antitumor Effector Mechanisms**

Cell-mediated immunity is the dominant antitumor mechanism *in vivo*. Although antibodies can be made against tumors, there is no evidence that they play a protective role under physiologic conditions.

- ❖ ***Cytotoxic T Lymphocytes***

cytotoxic T lymphocytes (CTLs) are responsible for the major immune defense mechanism against tumors. They seem to play a protective role, chiefly against virus-associated neoplasms (e.g., EBV-induced Burkitt lymphoma, HPV-induced tumors). The presence of MHC-restricted CD8⁺ cells that can kill autologous tumor cells within human tumors suggests that the role of T cells in immunity against human tumors may be broader than was previously suspected.

- ❖ ***Natural Killer Cells***

NK cells are lymphocytes that are capable of destroying tumor cells **without previous sensitization**; they may provide the first line of defense against tumor cells. After activation with IL-2, NK cells can lyse a wide range of human tumors, including many that seem to be nonimmunogenic for T cells. **T cells and NK cells apparently provide complementary antitumor mechanisms.** Tumors that fail to express MHC class I antigens cannot be recognized by T cells, but these tumors may trigger NK cells because the latter are inhibited by recognition of normal autologous class I molecules.

- ❖ ***Macrophages***

T cells, NK cells, and macrophages may collaborate in antitumor reactivity, because interferon- γ , a cytokine secreted by T cells and NK cells, is a potent activator of macrophages. Activated macrophages may kill tumors by mechanisms similar to those used to kill microbes (e.g., production of reactive oxygen metabolites) or by secretion of tumor necrosis factor (TNF).

- ❖ ***Humoral Mechanisms***

Although there is no evidence for the protective effects of antitumor antibodies against spontaneous tumors, administration of monoclonal antibodies against tumor cells can be therapeutically effective. A monoclonal antibody against CD20, a B cell surface antigen, is widely used for treatment of certain non-Hodgkin lymphomas.

- ❖ **Immune Surveillance and Immune Evasion by Tumors**

- ✓ The strongest argument for the existence of immune surveillance is the increased frequency of cancers in immunodeficient hosts. But again, cancers also occur in persons who do not suffer

from any overt immunodeficiency. And hence, there must be a mechanism by which cancers evade the immune system in immune-competent hosts. Several escape mechanisms have been proposed:

- *Selective outgrowth of antigen-negative variants.* During tumor progression, strongly immunogenic sub-clones may be eliminated.

- *Loss or reduced expression of histocompatibility molecules.* Tumor cells may fail to express normal levels of human leukocyte antigen (HLA) class I, escaping attack by CTLs. Such cells, however, may trigger NK cells.
- *Immunosuppression.* Many oncogenic agents (e.g., chemicals, ionizing radiation) suppress host immune responses. Tumors or tumor products also may be immunosuppressive. For example, TGF β , secreted in large quantities by many tumors, is a potent immunosuppressant. In some cases, the immune response induced by the tumor may inhibit tumor immunity. Several mechanisms of such inhibition have been described. For instance, recognition of tumor cells may lead to engagement of the T cell inhibitory receptor, CTLA-4, or activation of regulatory T cells that suppress immune responses.
- *Antigen masking.* Many tumor cells produce a thicker coat of external glycocalyx molecules, such as sialic acid-containing mucopolysaccharides, than normal cells. This thick coat may block access of immune cells to antigen-presenting molecules, thereby preventing antigen recognition and cell killing.
- *Downregulation of co-stimulatory molecules.* Costimulatory molecules are required to initiate strong T cell responses. Many tumors reduce expression of these costimulatory molecules.

• **Clinical features of neoplasia**

- I. Local effects
 - compression of adjacent structures**
 - Mechanical obstruction** - particularly those projecting to hollow structures like colonic tumors which may cause intestinal obstruction or those projecting to the gastric lumen causing gastric outlet obstruction.
 - Infarction & hemorrhage**—esp for tumors which tend to move around having a stalk, particularly ovarian tumors having a stalk tend to undergo torsion on the stalk and this may cause hemorrhage & eventually result in infarction.
 - Ulceration** – particularly fungating tumors (benign or malignant), Example, A giant fibro adenoma of the breast assume a very large size and form surface ulceration. If a malignant tumor form surface ulceration it likely will bleed and cause anemia.
- II. **Cancer Cachexia** –progressive wasting characterized by marked loss of adipose tissue and skeletal muscle. 2 factors are known to operate (tumor factors and host factors). Factors produced by tumors include protein proteolysis inducing factor (bring destruction or catabolism of the skeletal muscle) and tumor lipid mobilizing factor (cause lipolysis). While the host factors include cytokines produced by inflammatory cells which increase in Acute phase proteins produced by the liver and are associated with cachexia. The cytokines are responsible for the wasting by stimulating production of these proteins instead of albumin. Acute phase proteins are responsible for mobilization of proteins from skeletal muscle and triglyceride from adipose tissue. Cachexia is also known as Anorexia.
- III. Fever of unknown origin

- IV. **Tumor lysis Syndrome** – characterized by excessive destruction of cancer cells (particularly those having high proportion of growth factors) – malignant lymphomas and leukemia have high growth fractions so respond to chemotherapy or radiation undergoing high amount of tumor lysis and giving cancer cell products which may have effect in the host.
- V. **Paraneoplastic Syndromes**: a group of conditions developing in patients with advanced cancer which are neither explained by direct and distant spread of the tumor, nor by the usual hormone elaboration by the tissue of origin of the tumor. A good example can be ectopic hormone production.

❖ The coming 8 conditions are paraneoplastic syndromes.

1. **Endocrine syndrome**

Hypercalcaemia – the most frequent paraneoplastic syndrome. Commonly produced by **bronchogenic squamous cell carcinoma** producing ectopic parathyroid hormone related protein (PTHrp)

Cushing Syndrome – **Small cell carcinoma of the lung** is known to produce ectopic ACTH (Adrenocorticotrophic hormone) resulting in hyperplasia and hypertrophy of the adrenal cortex

Polycythemia – particularly in **hepatocellular carcinoma/hepatoma**. Liver cells normally do not produce erythropoietin but they do in these cases. **Pancreatic carcinomas** also produce erythropoietin.

Hypoglycemia – **commonly seen in hepatoma and hepatocellular carcinoma**. Cancer cells produce insulin like factors causing hypoglycemia.

2. **Neuromyopathic syndromes** – includes **peripheral neuropathies** and progressive loss of neurons from the CNS associated with destruction of the muscles, **cortical cerebellar degeneration (degeneration of purkinje cells by immunological damage**, not explained with the presence of local malignant or metastatic tumors this has something to do with immunological damage), several myopathies and **poly myositis, myasthenia gravis syndrome** as a result to damage of neurons.

3. **Effects on osseous (bone), joints and soft tissue**

clubbing of fingers – swelling and raising of nails of the digits usually associated with enlargement of inter-phalangeal joints because of an increase in synovial tissue, synovial membrane and even blood vessels around the joints.

hypertrophic osteoarthropathy – usually seen in **bronchogenic carcinomas**. Thickening of the skin and bone because there is periostitis (result in reactive laying down of the bone which accounts for bone thickening). There is also arthropathy (increase in synovial membrane, synovial membrane projected in to villus like structures associated with an increase in vascular connective tissue)

4. **Hematologic and vascular syndromes**

Venous thrombosis – Common in patients with advanced cancer. This is usually in the form of migratory thrombophlebitis/white leg (vein inflammation caused by thrombosis). Commonly observed in pancreatic carcinoma and carcinomas in GI tract.

DIC (Disseminated intravascular coagulation) -

Leukemoid reaction (increase in WBC count mimicking leukemia)

Thrombotic endocarditis

Anemia – anemia of chronic illness

5. Gastrointestinal syndromes – patients may be present with nausea, vomiting and malabsorption syndrome not explainable with the presence of local or metastatic cancer.

6. Renal syndromes

renal vein thrombosis

amyloidosis (can also exist in other organs)

7. Cutaneous syndromes (skin manifestation in patients with advanced cancer)

Acanthosis nigricans – presence of dark brown verrucous skin lesions. May be observed in the mucous membrane as well. Increased melanin pigmentation in the basal cells because of an increase in squamous epithelium (hyperplastic) covered with very thick keratin. Commonly seen around the **axilla, groin**

Seborrheic dermatitis – Increased dark brown verrucous structures.

Exfoliative dermatitis – reddening of the skin

8. Amyloidosis – is an acellular fibrillary protein (amyloid) deposited extracellularly leading to enlargement of the organs.

• **Common carcinogenic agents**

1. **Chemical carcinogen**-Any discrete chemical compound which has been shown to cause cancer in human or animal studies by either initiating or promoting cancer. Most agents are mutagenic

✓ **Direct-Acting Agents**-require no metabolic conversion and are in general weak carcinogens but are important because some of them are cancer chemotherapy drugs (e.g., alkylating agents) used in regimens that may cure certain types of cancer (e.g., Hodgkin lymphoma), only to evoke a subsequent, second form of cancer, usually leukemia.

✓ **Indirect-Acting Agents**-require metabolic conversion to a carcinogen. One of the most potent indirect chemical carcinogens are

I. Polycyclic hydrocarbons present in fossil fuels example, benzo pyrene and other carcinogens are formed in the high temperature combustion of tobacco in cigarette smoking. These products are implicated in the causation of lung cancer in cigarette smokers. Polycyclic hydrocarbons also may be produced from animal fats during the process of broiling meats and are present in smoked meats and fish. Epoxides, the principal active products

in many hydrocarbons which form covalent adducts (addition products) with molecules in the cell, principally DNA, but also with RNA and proteins.

- II. Aromatic amines and azo dyes
- III. Naturally occurring products of plant and microbial origin e.g. Aflatoxin B1 is of interest because it is a naturally occurring agent produced by some strains of *Aspergillus*, that grows on improperly stored grains and nuts. Has strong correlation incidence of hepatocellular carcinoma
- IV. Others-vinyl chloride, arsenic, nickel, chromium, insecticides, fungicides, polychlorinated biphenyls. Nitrites used as food preservatives have also caused concern, since they cause nitrosylation of amines contained in the food. The nitrosamines thus formed are suspected to be carcinogenic

2. **Physical carcinogens**-are classified in to radiation and non-radiation

- ✓ **Non-radiation**-soft or hard insoluble materials that include fibers, particulate matter, hard and soft synthetic materials and gels
- I. Fibers -Asbestos and other mineral fibers cause necrotic death of mesothelial cells with the consequent release of HMGB1(high mobility group protein B1) in the microenvironment which binds to mesothelial cells and macrophages, where it induces Nalp3 inflammasome. As a result, mesothelial cells and macrophages release pro-inflammatory cytokines including TNF- α and IL-1 β that cause chronic inflammation, followed by malignant transformation. In addition to mesothelioma asbestos also causes lung cancer via DNA damage by ROS and NOS and reduction of tumor immunity. It reduces the expression of chemokine receptor, CXCR3, and expression and production of IFN- γ with increasing IL-6 leading to chronic inflammation.

Other fibers--Erionite it is also associated with mesothelioma.

-Glass wool is associated with mesothelioma and lung cancer

- Manmade fibers include rock wool, slag wool and ceramic fibers. Of these, most of the evidence of carcinogenesis comes from animal studies.

- II. Particles-include metals such as cobalt and nickel and crystals such as silica. Like fibers, these are most often inhaled, but can also be ingested.
 - Inhaled crystalline silica produced lung cancer, and can produce lymphomas when injected into the chest or abdominal cavities of laboratory animals.
 - When injected into the subcutaneous tissue, metallic cobalt and metallic nickel produce sarcomas

- III. Hard and soft materials-metallic alloys used in joint replacement surgery and synthetic plastics used in vascular surgery the most common type of tumor seen is a sarcoma.

-Silicon gel, used in breast implants, is another potential physical carcinogen. It produces fibro sarcomas and plasmacytomas. Others- gold,platinum, silver, steel and a variety of different plastics.

- ✓ **Radiation**-whatever its source (UV rays of sunlight, x-rays, nuclear fission, radionuclides) is an established carcinogen.

- a. Non ionizing-radiations never create electrically charged IONS (cations⁺ and anions⁻) when tar-geting atoms or molecules.eg.UV induce the formation of pyrimidine dimers within DNA, leading to mutations. Therefore, UV rays can give rise to squamous cell carcinomas and melanomas of the skin.

UVA- long wave, "black-tanning light"

UVB- medium wave, "inflammatory"

UVC- short wave, "killing germs"

- b. Ionizing radiations-For direct action, damage occurs as a result of ionization of atoms on key molecules in the biological system. This causes inactivation or functional alteration of the molecule. Indirect action involves the production of reactive free radicals whose toxic damage to the key molecule has a biological effect

- There are four stages in action of ionizing radiation on biological structures: physical, physic-chemical, chemical and biological stages.
- On physical stage the transfer of kinetic energy from ionizing radiation to atoms or molecules leads to excitation and ionization of these atoms or molecules.
- On physic-chemical stage the displace of absorbed energy of ionizing radiation into molecules and between them and formation of free radicals take place.
- On chemical stage reactions between free radicals, reactions between free radicals and intact molecules, formation of molecules with abnormal structure and function realize.
- There are injuries on all levels – from cellular structures to organism and population, development of processes of biological damage and reparative processes occur on biological stage.

3. Oncogenic microbes and viruses

- ✓ Oncogenic RNA Viruses- only one retrovirus, the human T cell lympho tropic virus1 (HTLV-1), has been demonstrated to cause cancer in humans.HTLV-1 is associated with a form of T cell leukemia/lymphoma. HTLV-1 has tropism for CD4⁺T cells, making them the major target for neoplastic transformation.Human

infection requires transmission of infected T cells through sexual intercourse, blood products, or breastfeeding. Leukemia develops only in about 3% to 5% of infected persons after a long latent period of 20 to 50 years. The molecular mechanisms of transformation are not clear as it doesn't contain viral oncogenes. A unique region called pX which contains several genes, including one called TAX. The TAX protein has been shown to be necessary and sufficient for cellular transformation. TAX based inactivation of tumor suppressor genes such as TP53, the proliferating T cells are at increased risk for secondary transforming events (mutations), which lead ultimately to the outgrowth of a monoclonal neoplastic T cell population.

- ✓ Oncogenic DNA Viruses- Four DNA viruses—HPV, Epstein Barr virus (EBV), Kaposi sarcoma herpesvirus (KSHV, also called human herpesvirus-8 [HHV-8]), and hepatitis B virus (HBV)—are of special interest because they are strongly associated with human cancer.
 - Low risk HPVs, predominantly HPV-6 and HPV-11 are associated with genital warts. By contrast, high risk HPVs (e.g., types 16 and 18) cause several cancers, particularly squamous cell carcinoma of the cervix and anogenital region. High-risk HPV types simulates the loss of tumor suppressor genes, activates cyclins, inhibits apoptosis, and combats cellular senescence. The oncogenicity of HPV is related to the expression of two viral oncoproteins, E6 and E7; they bind to Rb and p53, respectively, neutralizing their function.
 - EBV is associated with, Burkitt lymphoma and has been discovered with the diverse list of tumors, including B cell lymphomas in patients with defective T cell immunity, a subset of Hodgkin lymphoma, nasopharyngeal carcinoma, a subset of T cell lymphomas, gastric carcinomas, NK cell lymphomas, and even, in rare instances, sarcomas, mainly in the immunosuppressed
 - EBV uses the complement receptor CD21 to attach to and infect B cells such infection leads to polyclonal B cell proliferation and generation of B lymphoblastoid cell lines.
 - EBV encoded genes, called LMP1 (latent membrane protein 1) acts as an oncogene. LMP1 prevents apoptosis by activating BCL2. Thus, the virus “borrows” a normal B cell activation pathway to promote its own replication by expanding the pool of cells susceptible to infection
 - Hepatitis B and Hepatitis C Viruses-evidence linking chronic HBV and hepatitis C virus (HCV) infection with hepatocellular carcinoma is strong. It is estimated that 70% to 85% of hepatocellular carcinomas worldwide are due to infection with HBV or HCV. The HBV and HCV genomes do not encode any viral oncoproteins. The HBx protein of HBV and the HCV core protein can activate a variety of signal transduction pathways that also may contribute to carcinogenesis.

- *Helicobacter pylori*- *H. pylori* infection is implicated in the genesis of both gastric adenocarcinomas and gastric lymphomas. It involves increased epithelial cell proliferation on a background of chronic inflammation.
- The mechanism of *H. pylori*-induced gastric cancers is multifactorial, including immunologically mediated chronic inflammation, stimulation of gastric cell proliferation, and production of reactive oxygen species that damage DNA. *H. pylori* pathogenicity genes, such as CagA, also may contribute by stimulating growth factor pathways.
- It is thought that *H. pylori* infection leads to polyclonal B cell proliferations and that eventually a monoclonal B cell tumor (MALT lymphoma) emerges as a result of accumulation of mutations.

• ANTINEOPLASTIC DRUGS

anticancer drugs calcified into three categories:

1. Class 1 agents (e.g., radiation, mechlorethamine, and carmustine)

- exert their cytotoxicity in a *nonspecific* manner.
- They kill both normal and malignant cells to the same extent.

2. Class 2 agents are phase specific

Only a certain *proportion* of cells are sensitive to the toxic effects of these drugs.

For example-hydroxyurea and cytarabine kill only cells in the S-phase.

-bleomycin is most toxic to cells in G2- and early M-phases

3. Class 3 agents are proliferation-dependent but non-phase-specific agents

ALKYLATING AGENTS

General MOA

transfer of their alkyl groups to various cellular constituents nitrogen (N7) and 6-oxygen (O6) of guanine particularly.

These interactions can occur on A single strand or- Both strands of DNA. major alkylating agents are bifunctional with two reactive groups

There are several possible consequences of N7 guanine alkylation:

1. Cross-linkage. Bifunctional alkylating agents, such as the nitrogen mustards, may form covalent bonds with each of two adjacent guanine residues. Such interstrand cross-linkages will *inhibit DNA replication and transcription*. Intrastrand cross-links also may be produced between DNA and a nearby protein.

2. Mispairing of bases. Alkylating at N7 changes the O6 of guanine to its enol tautomer, which can then form base pairs with thymine. This may lead to gene miscoding, with adenine- thymine pairs replacing guanine-cytosine. The result is the production of *defective proteins*.

3. Depurination. N7 alkylation may cause cleavage of the imidazole ring and excision of the guanine residue, leading to *DNA strand beakage*.

❖ *Cross-linking of DNA appears to be of major importance to the cytotoxic action.*

- ❖ alkylating agents are not cell cycle specific but Cells are most susceptible to alkylation in late G1 and S phases of the cell cycle

Mechanisms of acquired resistance

- ↑ capability to repair DNA lesions
- ↓ permeability of the cell to the alkylating drug
- ↑ production of glutathione (Inactivates the alkylating agent through conjugation)
- ↑ glutathione S-transferase activity (Catalyzes the conjugation)

NITROSOUREAS (Carmustine, Lomustine, streptozocin and Semustine)

Highly lipid-soluble and are able to cross the BBB so useful in brain tumor.

Streptozocin has minimal bone marrow toxicity.

Platinum Analogs (Cisplatin, Carboplatin, Oxaliplatin)

Kills cells in all stages of the cell cycle by formation of intrastrand and interstrand cross-links, and inhibits DNA synthesis and function.

- primary binding site is the N7 POSITION OF GUANINE
 - ❖ *Aggressive hydration with intravenous saline infusion significantly reduce the incidence of nephrotoxicity.*
 - ❖ Amifostine : antidote for cisplatin (some alkylating agents) nephrotoxicity

Carboplatin

- Cytotoxic effects exactly as cisplatin & has activity against the same spectrum of solid tumors.
- Has significantly less renal toxicity & GI toxicity than cisplatin.
- IV hydration is not required
- Has now widely replaced cisplatin

ANTIMETABOLITE

The antimetabolites are structurally similar to endogenous compounds and are antagonists of folic acid, purines or pyrimidines.

Antimetabolites are CCS drugs acting primarily in the S phase of the cell cycle.

ANTAGONIST OF FOLIC ACID

Methotrexate

MOA

- inhibit Dihydrofolate reductase which interferes with the synthesis of the reduced form that accepts one-carbon units. This action leads to a decrease in the synthesis of thymidylate, purine nucleotides, and amino acids and thus interferes with nucleic acid and protein metabolism.

The formation of polyglutamate derivatives of methotrexate appears to be important for cytotoxic actions via selectively retaining methotrexate within cancer cell.

Mechanism of resistance

- ↓ drug transport
- ↓ formation of cytotoxic MTX polyglutamates,
- Synthesis of increased levels of DHFR through gene amplification
- Altered DHFR with reduced affinity for MTX.

Pharmacokinetics—

Methotrexate is not metabolized, and its clearance is dependent on renal function. Adequate hydration is needed to prevent crystallization in renal tubules.

Toxicity

The toxic effects of methotrexate on normal cells may be reduced by administration of folinic acid (leucovorin); this strategy is called **leucovorin rescue**.

Purine antagonist ----- (6-thiopurine)

Mercaptopurine (6-MP) and Thioguanine (6-TG)

MOA

Inactive in parent drug form

Both drugs are activated by hypoxanthine-guanine phosphoribosyl transferases (HGPRTases) to toxic nucleotides that inhibit several enzymes involved in purine metabolism

Mechanism of drug resistance

-Decrease activity of HGPRTase,

- increase their production of alkaline phosphatases that inactivate the toxic nucleotides.

Pharmacokinetics—Mercaptopurine and thioguanine have low oral bioavailability because of first-pass metabolism by hepatic enzymes.

6-MP converted to an inactive metabolite (6-thiouric acid) by xanthine oxidase. **but 6-TG is not substrate for xanthine oxidase.** xanthine oxidase inhibitors allopurinol and febuxostat inhibit

metabolism of 6-MP. therefore dose of 6-MP must be reduced when allopurinol is

Simultaneously used.

❖ 6-MP Used primarily in the treatment of childhood acute leukemia.

PURINE ANALOG

Fludarabine –PRODRUG

Phosphorylated intracellularly to the triphosphate form can directly incorporated into DNA Resulting in inhibition of DNA synthesis and function or may inhibit DNA polymerase α and β which interfere with DNA repair.

PYRIMIDINE ANALOG

A.FLUOROURACIL (5-FU)

MOA

PRODRUG-5-FU is converted to 5-FdUMP, which competes with deoxyuridine monophosphate (dUMP) for the enzyme thymidylate synthetase. Forms a covalently bound ternary complex with Thymidylate synthase and the reduced folate *N*5,10-Methylenetetrahydrofolate, a reaction critical for the de novo synthesis of thymidylate. This results in inhibition of DNA synthesis through "Thymineless" death. Incorporation of FdUMP into DNA inhibits DNA synthesis and function while incorporation of 5-fluorouridine- 5'-triphosphate (FUTP), another 5-FU metabolite, into RNA interferes with RNA processing and function. Therefore, cytotoxic action is both DNA and RNA mediated.

Pharmacokinetics—When given intravenously, fluorouracil is widely distributed, including into the cerebrospinal fluid. Elimination is mainly by metabolism.

CLINICAL USE

- ❖ 5-FU- the most widely used agent in the treatment of colorectal cancer.

B.capecitabine

MOA: After being absorbed, capecitabine, which is itself nontoxic, undergoes a series of enzymatic reactions, the last of which is hydrolysis to 5-FU. This step is catalyzed by thymidine phosphorylase an enzyme that is concentrated primarily in tumors.

Clinical use

Mainly Used in the treatment of metastatic breast Cancer.

PYRIMIDINE ANTAGONIST

CYTARABINE (cytosine arabinoside)

MOA

Cytarabine is a pyrimidine antimetabolite. The drug is activated by kinases to AraCTP, an inhibitor of DNA polymerases α and β which result in blockage of DNA synthesis and DNA Repair, respectively.

MOST S phase-specific antimetabolite

Clinical use

Activity limited exclusively to hematologic malignancies,

Has absolutely no activity in solid tumors.

NATURAL PRODUCT ANTICANCER

DRUGS

INCLUDE plant-derived CCS drugs

- Vinca alkaloids (**vinblastine, vincristine, vinorelbine**),
- podophyllotoxins(**etoposide, teniposide**),
- camptothecins (**topotecan, irinotecan**),
- taxanes (**paclitaxel, docetaxel**).

1. VINA ALKALOIDS

MOA—The vinca alkaloids block the formation of the mitotic spindle by preventing the assembly of tubulin dimers into microtubules.

Act primarily in the M phase of the cancer cell cycle.

MECHANISM OF RESISTANCE – increased efflux of the drugs from tumor cells via the membrane drug transporter.

Pharmacokinetics—These drugs must be given parenterally. They penetrate most tissues except the cerebrospinal fluid. They are cleared mainly via biliary excretion.

CLINICAL USE

Vincristine used with prednisone effectively combined with prednisone For remission induction in acute lymphoblastic leukemia in children.

2.EPIPODOPHYLLOTOXINS

• VP-16 (ETOPOSIDE) & VM-26 (TENIPOSIDE)

- Are semisynthetic derivatives of podophyllotoxin.
- Block cell division in the late S-G2 phase of the cell cycle.

– Inhibition of topoisomerase II, Results in DNA damage through strand breakage induced by the formation of a ternary complex of drug, DNA, and enzyme.

CLINICAL USE

ETOPOSIDE have wide clinically for many tumors.

TENIPOSIDE Use is limited mainly to acute lymphoblastic leukemia.

ANTITUMOR ANTIBIOTICS

1. Anthracyclines

GENERAL MECHANISM OF ACTION

1. Inhibition of topoisomerase II
2. Intercalation DNA
3. Membrane disruption result in change in fluid and ion transport.
4. Generate semiquinone and oxygen free radicals

these cause cardiotoxicity associated side effect *Dexrazoxane*, an inhibitor of iron-mediated free radical generation, may protect against the dose-dependent form of cardiotoxicity.

Group Resistance :- via an increased efflux of the antibiotic from the cell via P-glycoprotein.

A. Dactinomycin

MOA:- tightly to double-stranded DNA CAUSE intercalation between adjacent G-C base pairs.

- Inhibits all forms of DNA-dependent RNA synthesis.

CLINICAL USE

Mainly used to treat pediatric tumors (e.g Wilms' tumor, rhabdomyosarcoma, and Ewing's sarcoma,)

B. Doxorubicin

MOA:- 2,3 and 4 mentioned above as general mechanism of anthracyclines.

CLINICAL USE in combination with other agents for treatment of sarcomas and a variety of Carcinomas. Response and Remission tend to be improved.

2. MITOMYCIN

MOA :-undergoes metabolic activation Through enzyme-mediated reduction To generate alkylating agent that cross-links DNA.

Active in all stages of cell cycle.

CLINICAL USE

Hypoxic tumor stem cells of solid tumors are more sensitive to its cytotoxic actions. mainly used in treatment of squamous cell carcinoma of anus.

3. Bleomycin

Moa :- small peptide Contains a DNA-binding region & an iron-binding domain at opposite ends of the molecule. Acts by binding to DNA Results in single-strand & double-strand breaks Following free radical formation, and inhibition of DNA biosynthesis.

Act specifically at **G2 stage** of cell cycle.

Clinical use:- widely used in treatment of different tumors.

Limiting adverse effect: Cause pulmonary fibrosis.

HORMONAL AGENTS

A. Gonadal Hormone Antagonist

TAMIOXFINE

MOA:-A competitive partial agonist-inhibitor of estrogen

Clinical use :- in receptor positive breast carcinoma and has been shown to have a preventive effect in women at high risk for breast cancer.

Adverse effect:- Because it has agonist activity in the endometrium, tamoxifen increases the risk of endometrial hyperplasia and neoplasia.

2. Glucocorticoids

Prednisone is the most commonly used glucocorticoid in cancer chemotherapy and is widely used in combination therapy for leukemias and lymphoma

MISCELLANEOUS ANTICANCER AGENTS

HYDROXYUREA– An analog of urea

– MOA involves the inhibition of DNA synthesis

- By inhibiting the enzyme ribonucleotide reductase,

– Exerts its effects in the S phase of the cell cycle.

Clinical use

- Chronic myelogenous leukemia and
- Treatment of acute myeloid leukemia

Tyrosine Kinase Inhibitors

Imatinib

MOA :-inhibits the tyrosine kinase activity of the protein product of the *bcr-abl* oncogene that is commonly expressed in chronic myelogenous leukemia (CML). In addition imatinib is effective for treatment of gastrointestinal stromal tumors that express the *c-kit* tyrosine kinase, which is also inhibited.

Growth Factor Receptor Inhibitors

Bevacizumab is a monoclonal antibody that binds to **vascular endothelial growth factor (VEGF)** and prevents it from interacting with VEGF receptors. VEGF plays a critical role in the angiogenesis required for tumor metastasis. Can be safely & effectively combined with other antineoplastic drug in the treatment of metastatic colorectal cancer.

Cetuximab is a chimeric monoclonal antibody AGAINST EGFR. It is used in combination with other drugs for metastatic colon cancer and is used in combination with radiation for head and neck cancer.

Panitumumab is a fully human monoclonal antibody directed against the EGFR; it is approved for refractory metastatic colorectal cancer.

Gefitinib and **erlotinib** are smallmolecule inhibitors of the EGFR's tyrosine kinase domain. Both are used as second-line agents for non-small cell lung cancer, and erlotinib is also used in combination therapy of advanced pancreatic cancer.

Common Toxicities of Antineoplastic Agents

1. MUCOSITIS

- GI mucosa composed of epithelial cells with a high mitotic index and rapid turnover rate,
- Common site of chemotherapy-induced toxicity

2. ALOPECIA

3. INFERTILITY

4. MYELOSUPPRESSION

– The most common-limiting side effect of cytotoxic agents– WBCs, especially neutrophil precursors, are most significantly affected than platelets than erythrocytes

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